Development of a Mathematical Model for the Simulation of Ex Vivo Ocular Tissue Clearing

Proof of “Heat flux from CUBIC concentration” Equation

Proof. To model the flow of heat transfer, Fourier’s Law of Heat Convection was used for the foundation,

$$q = -K_{\text{eff}} \nabla T$$

where:

$$K_{\text{eff}} = \varepsilon K_s + (1 - \varepsilon) K_p + K_{\text{disp}}.$$

Due to its insigificance in the protocol, K_{disp} was removed because diffusion of the solutions occurs through passive diffusion, which involves slow fluid movement.

To represent how diffusion changes from temperature fluctuations, a diffusion function is derived from Brownian Motion’s Formula and Bruggeman’s Model,

$$D(\varepsilon) = D_0 \varepsilon^{\frac{3}{2}}$$

which is coupled with the previously derived change in pore pressure equation to model the influence of pore diameter:

$$-D(\varepsilon) \frac{\partial p}{\partial t}.$$

For the thermal conduction component of the equation, the thermal conductivity is replaced by a constant value, resulting in the full derived equation:

$$\boxed{q = [-D(\varepsilon) \nabla p - (1 - \varepsilon) 0.565] \nabla T}.$$